

Sushant Nagil

DevOps, Site Reliability & Cloud Infrastructure Engineer

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Summary

DevOps engineer with 3 years automating, deploying and supporting large-scale, multi-cloud AWS and Oracle Cloud (OCI) infrastructure for India's state electricity utilities. Lead day-to-day work for a five-person infrastructure team, mentor junior engineers, and communicate directly with application teams, utilities and cloud vendors. Across 22 AWS accounts and four production Amazon EKS clusters: Kubernetes autoscaling, Kafka and MQTT distributed systems on Linux, Terraform, CI/CD, Python automation, and FinOps cost optimization worth about \$45K a year. Certified: CKA, AWS Solutions Architect, AWS Data Engineer, Terraform Associate.

Experience

Polaris Smart Metering

May 2025 – Present

DevOps Engineer

Jaipur, India

- Lead day-to-day work for the five-person infrastructure team and mentor junior DevOps engineers and the security and support teams: scope the Jira stories, run sprint planning and design reviews, review every change to shared production accounts, and act as the infrastructure contact for application teams, state utilities and cloud vendors, from integration issues to pricing.
- Lead on-call, incident troubleshooting and postmortems. Root causes I debugged: PostgreSQL connection failures from Linux strict memory overcommit, and Kafka consumer lag after VPA cut decoder CPU.
- Own FinOps across 22 AWS accounts: about \$45K/yr in recurring savings from rightsizing, removing an idle node group, CloudWatch agent cleanup across 6 accounts and 100% Aurora Reserved Instance coverage. Set ECR lifecycle rules on 500+ repositories and removed 167 stale pipelines, build projects and registries, backing up each batch first. Also moved a legacy Python 2.7 and MySQL 8 application to smaller infrastructure, from \$2,200 to \$1,300 a month.
- Led the migration of our Kafka clusters, a distributed system at the core of the platform, from Amazon MSK to Strimzi on EKS. When older consumers using lz4 compression fell behind during the cutover, wrote a small transcoder so they kept consuming, and the migration finished without a rollback.
- Designed and run layered autoscaling for the head-end and meter-data microservices across four production EKS clusters: Karpenter for nodes, KEDA on Kafka lag, HPA for request traffic and VPA for right-sizing (with Goldilocks). Moved workloads from managed node groups to Karpenter, keeping steady meter-data services on fixed capacity.
- Cut typical build time from 30 minutes to about 3 by reworking a CI/CD pipeline that rebuilt the whole platform on every commit; it now builds only the services whose paths changed and deploys across accounts with shared IAM roles.
- Building a new state's production platform as code in Terraform, with dev, stage and prod environments and Argo CD for deployments.
- Led the EKS upgrade from 1.33 to 1.35 across five AWS accounts before extended-support pricing began (custom AMIs, launch templates, instance metadata settings).
- Own the AWS Organizations landing zone and network: OU layout, service control policy (SCP) security and compliance guardrails, tag policy (tagged instances up from 56% to 100%), IAM Identity Center (SSO) permission sets, per-tier VPCs and site-to-site VPN into on-prem data centres.
- Designed PostgreSQL disaster recovery (DR) on EC2 with pgBackRest backups to delete-proof S3 instead of streaming replicas, keeping RTO and RPO predictable without doubling compute.
- Built a capacity model for a 10-million-meter rollout from 14 days of production data, showing that pods and Kafka partitions scale out and the write-heavy database writer does not.

Polaris Smart Metering

Nov 2023 – Apr 2025

AWS Cloud Engineer

Jaipur, India

- Designed the IAM and AWS SSO (IAM Identity Center) model across the accounts, with least-privilege roles and per-team permission sets, and moved the team off standalone IAM users.

- Ran the AWS Migration Acceleration Program (MAP) assessment and wave plan, then moved Redis from EC2 to ElastiCache and Kafka from EC2 to MSK, which removed two of the biggest sources of on-call pages.
- Stood up our first EMQX MQTT cluster, sized for 100,000+ concurrent meter connections feeding the head-end system.
- Automated dev and stage shutdowns with Lambda and EventBridge, cutting non-production AWS spend by ~40%, and built CodePipeline CI/CD for Lambda jobs that replaced manual deploys.
- Performance-tuned self-managed PostgreSQL on EC2 (pgtune, PgBouncer pooling) for the write-heavy metering workload, ran RabbitMQ clusters, and wrote the vacuum and restore runbooks the team still uses.
- Built the monitoring and observability stack for metrics, logs and alerts (Prometheus, Grafana, Loki, Fluent Bit, OpenTelemetry alongside CloudWatch) with SNS alerting to Slack and email for first-line on-call.

Polaris Smart Metering

Aug 2023 – Oct 2023

Cloud Engineering Intern

Jaipur, India

- Mapped the AWS footprint (accounts, VPCs, EC2, security groups) used in the later MAP assessment, set up the first CloudWatch alarms and cost-allocation tags, and converted to full-time.

Skills

- **Programming & Linux:** Python software development for automation and tooling, Bash; Linux/Unix systems administration and operating system troubleshooting; data structures & algorithms (learning)
- **Kubernetes (CKA):** Amazon EKS, Karpenter, KEDA, HPA/VPA, Goldilocks, Helm, Argo CD, Docker
- **AWS:** EC2, Lambda, EventBridge, S3 and EBS storage, IAM & IAM Identity Center, Organizations & SCPs, ECR, CloudWatch, RDS/Aurora, MSK, ElastiCache, WAF
- **Oracle Cloud (OCI):** Oracle Kubernetes Engine (OKE), VCN, DRG hub-and-spoke networking, Network Load Balancer
- **Networking & Security:** TCP/IP, VPC design, DNS (Route 53 private zones), load balancing (ALB/NLB), Nginx reverse proxy, firewalls (security groups, AWS WAF), SSL/TLS certificates (ACM) on load balancers and public domains, site-to-site VPN, VPC peering, IPv6
- **Infrastructure as Code & CI/CD:** Terraform (modules, remote state), CodePipeline, CodeBuild, GitHub Actions, Git/GitHub
- **Reliability, Monitoring & Observability:** On-call, incident response, postmortems, Prometheus, Grafana, Loki, Fluent Bit, OpenTelemetry, Zabbix
- **Data & Messaging:** PostgreSQL, pgBackRest, PgBouncer, MySQL, Redis, Apache Kafka, Strimzi, MirrorMaker 2, RabbitMQ, EMQX (MQTT)
- **FinOps:** Cost optimization, rightsizing, Savings Plans & Reserved Instances, ECR lifecycle, multi-account cost allocation

Certifications

- **AWS Certified Data Engineer, Associate:** May 2026
- **Certified Kubernetes Administrator (CKA):** The Linux Foundation, Dec 2025
- **HashiCorp Certified: Terraform Associate:** Sep 2025
- **AWS Certified Solutions Architect, Associate:** Jul 2025

Education

B K Birla Institute of Engineering & Technology
B.Tech, Computer Science

2020 – 2024
 Pilani, India

- National Cadet Corps (NCC): Under Officer, C Certificate holder.